

LESSON 2.1

EXPLORING SAMPLE SPACE



LESSON INTRODUCTION

MA.7.DP.2.1 Determine the sample space for a simple experiment.

LEARNING TARGETS

- I can determine all possible outcomes of a given scenario.
- I can determine whether or not a simple experiment is fair.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.DP.2.1 Determine the sample space for a repeated experiment.

ESSENTIAL QUESTIONS

- How does the number of possible outcomes for an experiment relate to the likelihood of a particular event?
- How can we apply probability to inform a prediction?

LESSON NARRATIVE

This lesson is the first within the unit, which introduces probability. The lesson engages students in thinking about sample space by using contexts they are already familiar with – games and fairness. While key vocabulary related to MA.7.DP.2.1 is introduced, this lesson is also designed to hook students by using their prior knowledge and experience relating to probability. In Lesson 2, students will delve more explicitly into determining sample spaces for simple experiments.

COMPONENT SUMMARY

2.1.1 WARM-UP (~5 MIN)

Estimation activity using benchmark percentage of 50%

2.1.3 EXPLORATION (5 MIN)

Students discover how “fairness” relates to probability

2.1.5 WRAP-UP (~5 MIN)

Students reflect on their understanding of the lesson

2.1.2 GUIDED INSTRUCTION (10 MIN)

Introduction of key probability terms

2.1.4 ACTIVITY (20-25 MIN)

Students compare sample spaces and fairness of carnival games in a gallery walk activity

RESOURCES

GLOSSARY

Key Terms:

- Outcome
- Sample space
- Event

Additional Terms:

- N/A

MATERIALS

- Gallery Walk images
- (Optional) Example game pieces to model gallery walk games

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may believe that all outcomes are equally likely in all scenarios.
- Students may believe that future outcomes in these experiments depend on previous outcomes (all examples in this lesson are of independent probability).
- Students may confuse the terms “outcome” and “event.”

THINGS TO CONSIDER

- In 2.1.3 Exploration, consider asking students to make the explicit connection between fractional sizes of the spinner sections and the “fairness” of the spinners.
- Model choosing a card from Doug’s chore list to demonstrate the difference between “outcome” and “event.” Using repetition can help students solidify the vocabulary.

LITERACY AND LANGUAGE ROUTINES

Number Talk

The 2.1.1 Warm-Up contains a question that asks students to estimate a percentage given a fraction and explain their reasoning. Allow each student time to independently come up with a strategy for estimating over or under 50%. Then, select students to share out different strategies they used in solving the problem. Consider adjusting the time for the Warm-Up to 10 minutes if utilizing this strategy.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

The 2.1.4 Activity is a collaboration opportunity for students. Students get to work together to review carnival game rules and determine whether or not those games should be considered “fair.” Student pairs also have the opportunity to discuss and reflect on their thinking with other student groups at the conclusion of the activity.

DIFFERENTIATE INSTRUCTION

The carnival games presented in 2.1.4 require students to read the rules of a game and determine the possible outcomes and “fairness” of that game. Auditory learners would benefit from hearing the verbal description of the activities and both kinesthetic and visual learners would benefit from watching a demonstration of the game.

2.1.1 WARM-UP: CAROLYN BEATRICE PARKER

TEACHER MOVES

Number Talk

Consider also allowing students to discuss their reasoning with a partner first before bringing the class back for a whole-group discussion. This practice can help students clarify their mathematical thinking.

ENRICHMENT

Do you know any schools in your area named after someone famous?



2.1.1 Warm-Up: Carolyn Beatrice Parker

Carolyn Beatrice Parker, born in Gainesville, FL in 1917, was the first African American woman known to receive a graduate degree in physics and subsequently work on the top-secret government project known as the Manhattan Project. Before earning her master's degree in physics, she earned a master's degree in mathematics and was also a teacher in Florida.



In 2020, an elementary school in Gainesville, FL was re-named in her honor. At this time, 11 of the 21 elementary schools in the Alachua County Public School system were named after people.

- Without performing any calculations, select the statement that best describes the percentage of the elementary schools in Alachua County that were named after people as of 2020.
 - significantly less than 50%
 - slightly less than 50%
 - exactly 50%
 - slightly more than 50%**
 - significantly more than 50%
- Explain your reasoning.
Answers will vary. The statement that best describes the percentage is slightly more than 50% because 21 is close to 20 and 10 is 50% of 20, therefore 11 is slightly more than 50% of 21.

2.1.2 GUIDED INSTRUCTION: DETERMINING OUTCOMES

ENRICHMENT

How do you find the total number of outcomes? Does one outcome seem more likely than another?

TEACHER MOVES

Students are becoming familiar with probability vocabulary for the first time. To have additional practice understanding the concept of sample space, consider prompting students to discuss the sample spaces for different experiments: outcomes of a Miami Heat game, shoes to wear to school, rolling a die, flipping a coin, etc.



2.1.2 Guided Instruction: Determining Outcomes

An **outcome** is a possible result of an experiment.

- A spinner divided into four equal areas is shown.

- How many outcomes are there from spinning the spinner?
4



In a probability model for a random process, the **sample space** is a list of the individual outcomes that are being considered.

- With your partner, discuss the sample space for the spinner. Write a list of the possible outcomes.
B, R, G, Y



An **event** is a set of possible outcomes resulting from an experiment. In general, an event is any subset of a sample space.

- How many outcomes are included in the event that the spinner lands on blue?
One
- Primary colors include yellow, blue, and red. How many outcomes are included in the event that the spinner lands on a primary color?
Three

2. Doug's parents created a set of cards with chores on them. Each Saturday, Doug chooses a card to determine his chore.

Mow the yard	Dust and vacuum the house	Clean the bathrooms	Clean your bedroom
Clean your bedroom	Mow the yard	Weed the garden	Mow the yard
Clean the bathrooms	Parents' choice	Dust and vacuum the house	Free day

- a. What are the possible outcomes for the chore cards?
Mow the yard, Dust and vacuum the house, Clean the bathrooms, Clean your bedroom, Weed the garden, Parents' choice, Free day
- b. Which outcomes have more than one card?
Mow the yard, Dust and vacuum the house, Clean the bathrooms, Clean your bedroom
- c. On a particularly hot Saturday, Doug is hoping he doesn't select an outdoor chore. Which outcomes could be included in the event that he chooses an outdoor chore?
Mow the yard, Weed the garden

- d. How many outcomes could be included in the event that Doug chooses an indoor chore?
6 - Dust and vacuum the house (2 cards), Clean the bathrooms (2 cards), and Clean your bedroom (2 cards)

3. Use the word bank to complete the summary of key concepts below.

event outcomes sample space set

To determine the sample space of a simple experiment, find all the outcomes that are possible and write them as a set.

An event is a subset of the sample space, such as rolling an even number on a 6-sided die.

2.1.2 GUIDED INSTRUCTION: DETERMINING OUTCOMES (CONTINUED)

TEACHER MOVES

Students should be able to complete question 2 independently. Consider having them check in with a partner to compare responses.

ENRICHMENT

Which outcome has the most cards? Which outcome(s) have the least cards?

DIFFERENTIATE INSTRUCTION

For English Language Learners, determining what constitutes an “outdoor” vs. “indoor” chore may be challenging to do independently. Consider providing a visual for each card as support for this activity.

DIFFERENTIATE INSTRUCTION

Consider asking the question, “if Doug chooses a card at random, which chore do you think he is most likely to get?” Students can begin to make connections between the number of outcomes and the probability of an event.

TEACHER MOVES

Clarify in question 3 that “set” and “list” can be used interchangeably in this context.

2.1.3 EXPLORATION: DIGGING DEEPER INTO OUTCOMES

TEACHER MOVES

Allow students to discuss which spinner is more favorable for each outcome. For example, pose the question, “If you wanted a spinner to land on red, which spinner would you prefer to use?” Encourage students to use mathematical support for their responses.

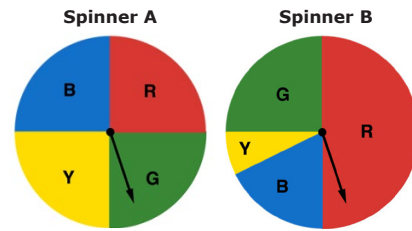
ENRICHMENT

- What does it mean for the spinner to be “fair?”
- Can you give an example of a fair game? An example of a game that is not fair?



2.1.3 Exploration: Digging Deeper into Outcomes

Two spinners are shown.



1. Work with your partner to compare and contrast the features of the two spinners.

Similarities	Differences
<i>size of the spinner</i> <i>amount of sections</i> <i>name and colors of the sections</i>	<i>size of the sections except for G</i>

2. One of the spinners is considered a fair spinner. Complete the sentence by selecting the fair spinner and writing an explanation.

- ☐ Spinner A
☐ Spinner B

is the fair spinner because all of the

sections have the same chance to be selected.



2.1.4 Activity: Gallery Walk

1. Work with your partner to determine the sample space for each game and whether the game is fair. Fill in the chart with your answers.

Game Name	Sample Space	Is it fair?
Skee-Ball	0, 10, 20, 30, 40, 50, 100	No
Card Draw	Clubs- Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King Spades- Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King Hearts- Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King Diamonds- Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King	Yes
Prize Wheel	orange, blue, red, yellow, white	No
Rubber Ducks	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25	Yes

2. Which game has the largest sample space?
Card Draw
3. Discuss with your partner which game's sample space was the most difficult to determine.
Answers will vary.

2.1.4 ACTIVITY: GALLERY WALK

DIFFERENTIATE INSTRUCTION

This activity requires students to read the rules for a game to determine whether or not the game is fair. For students who struggle with reading comprehension, it may be helpful to provide a demonstration of how each game is played. Consider splitting the class into four groups to simulate a few rounds of each game.

QUESTIONING

Consider asking students to make rule proposals to the games they deemed unfair. How can the rules be modified to make these games fair?

DIFFERENTIATE INSTRUCTION

Why do you think a carnival would choose to offer both fair and unfair games?

2.1.4 ACTIVITY: GALLERY WALK (CONTINUED)

TEACHER MOVES

Provide an opportunity for meaningful reflection with questions 3 and 4 on “fairness.” This response could be used for you to evaluate student understanding from the lesson.

4. Carnivals include many games similar to these. What do you notice about the fairness of such games?

Answers will vary. I notice that some games are noticeably fair or unfair, while other games are unnoticeably unfair.

5. Find another pair of partners to form a group of four. Compare your responses on whether each game is fair and discuss your reasoning. Record one similarity and one difference you notice in your reasoning.

Answers will vary.

Similarity	Difference
<i>We all thought that the Card Draw was fair because all of the suits had an equal chance of being drawn.</i>	<i>We thought that the Prize Wheel was not fair because all of the colors are not the same size, and our partners thought that the Prize Wheel was fair because all of the colors are the same size.</i>

6. Future lessons in this unit will explore probabilities with fair coins, fair dice, and fair spinners. Describe what you think it means for a coin, a die, or a spinner to be fair.

Answers will vary. Fair means that all of the outcomes have the same chance to be selected.

**2.1.5 Wrap Up: Reflection**

Complete each statement based on your understanding at this point.

Answers will vary.

1. I am still unsure about _____

_____.
2. One question I have that will help me better understand
is _____

_____?

2.1.5 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.